Unit 3.1 Printables: Age of the Earth

Every table students only copy or fill in is printed here, so they do not spend class time drawing it. Cut each piece on the dashed line and glue it into the matching "Glue in" box on that day's notebook setup page (Unit31_Student_Notebook.pdf). Print on white paper, black ink only.

Day	Printable	Use	Copies
Day 1	Unit CER page: question, claim, evidence log	Glue in · write on it	1 per student
Day 1	Frayer card sheet (homework)	Glue in · write on it	1 per student
Day 1	Hook data: ocean floor and continent ages	Glue in · reference	1 per student
Day 1	Site list for the plotted map	Glue in · write on it	1 per pair
Day 1	Plate boundary table	Glue in · write on it	1 per student
Day 2	Seafloor ages and color key	Glue in · reference	1 per student
Day 2	Seafloor strip	Glue in · write on it	1 per student
Day 2	Graph grid: age vs. distance	Glue in · write on it	1 per student
Day 2	North America crust ages and strip	Glue in · reference and write on it	1 per student
Day 3	Crater counts	Glue in · write on it	1 per student
Day 3	Timeline table and old materials	Glue in · reference and write on it	1 per student
Day 4	Guided notes	Glue in · write on it	1 per student
Day 4	Density table	Glue in · write on it	1 per student
Day 4	SketchNotes checklist	Glue in · write on it	1 per student
Day 5	Magnetic stripe bands	Glue in · reference	1 per student
Day 5	Hawaii and GPS	Glue in · reference and write on it	1 per student
Day 5	Fossil table	Glue in · reference	1 per student
Day 5	Evaluate the evidence	Glue in · write on it	1 per student
Day 6	Three buckets	Glue in · write on it	1 per student
Day 6	Part A sentence frames	Glue in · reference	1 per student
Day 6	Peer feedback	Glue in · write on it	1 per student
Day 6	Evidence cards	Glue in · write on it	1 per student
Day 6	Part B story frames	Glue in · write on it	1 per student
Day 7	Self-rating chart	Glue in · write on it	1 per student

Day 8 (CSA) needs nothing glued in. Things not printed here: the world map with latitude and longitude lines (Day 1), the continent cut-outs for the Pangaea puzzle (Day 5), and the 45 cm adding-machine tape strips (Day 3).

UNIT 3.1 · DAY 1 · GLUE IN · WRITE ON IT

Unit CER page: question, claim, evidence log

Your first claim goes under the question. Add a row each day. Row 1 is done for you. Glue this on its own new page.

Question: Why is the seafloor so much younger than the continents?

My first claim (Day 1):

Day	Evidence (use a number)	What it shows
1	The oldest ocean floor is about 200 million years old. Continental rock is up to 4.4 billion years old.	The seafloor is much younger than the continents.

My claim after Day 6:

UNIT 3.1 · DAY 1 · GLUE IN · WRITE ON IT

Frayer card sheet (homework)

For each word write a definition in your own words, draw a quick picture, and give an example. Row 1 is done for you. Finish by Day 7.

Word	Definition in my own words	Picture	Example
lithosphere (plate)	A rigid slab of Earth's outer layer that moves.		The Pacific plate
plate boundary (divergent, convergent, transform)			
mid-ocean ridge			
subduction			
trench			
magnetic reversal			
craton (a continent's ancient core)			
impact crater			

UNIT 3.1 · DAY 1 · GLUE IN · REFERENCE

Hook data: ocean floor and continent ages

Place on the ocean floor	Age of the rock
Right at the Mid-Atlantic Ridge	0 (forming now)
Middle of the Atlantic, about 2,000 km from the ridge	about 80 million years
Atlantic seafloor next to the east coast of North America	about 180 million years
Right at the East Pacific Rise	0 (forming now)
Western Pacific, east of Japan (the oldest on Earth)	about 180 million years

Continental material	Where	Age
Oldest mineral (a zircon crystal)	Jack Hills, Western Australia	about 4.4 billion years
Oldest known rock (Acasta Gneiss)	Northwest Territories, Canada	about 4.0 billion years
Very old rock (Isua belt)	Greenland	about 3.8 billion years
Oldest widely accepted fossils (stromatolites)	Western Australia	about 3.5 billion years

UNIT 3.1 · DAY 1 · GLUE IN · WRITE ON IT

Site list for the plotted map

Plot each site on your world map: a red dot for a volcano, a blue X for an earthquake. Each partner plots five. Check Plotted when done. Mount St. Helens is done for you.

Site	Type	Latitude	Longitude	Plotted
Mount St. Helens, Washington, USA	volcano	46.2° N	122.2° W	✓
Mount Fuji, Japan	volcano	35.4° N	138.7° E	
Popocatépetl, Mexico	volcano	19.0° N	98.6° W	
Hekla, Iceland	volcano	64.0° N	19.7° W	
Erta Ale, Ethiopia	volcano	13.6° N	40.7° E	
Tohoku, Japan (2011, magnitude 9.1)	earthquake	38.3° N	142.4° E	
Maule, Chile (2010, magnitude 8.8)	earthquake	35.9° S	72.7° W	
Haiti (2010, magnitude 7.0)	earthquake	18.4° N	72.5° W	
Turkey (2023, magnitude 7.8)	earthquake	37.2° N	37.0° E	
Kīlauea, Hawai‘i, USA	volcano	19.4° N	155.3° W	

UNIT 3.1 · DAY 1 · GLUE IN · WRITE ON IT

Plate boundary table

Fill in the table after you make each clay model. Divergent is done for you.

Boundary	What the plates did	What formed	Real example
Divergent	Moved apart	New crust at a mid-ocean ridge	Mid-Atlantic Ridge
Convergent			
Transform			

UNIT 3.1 · DAY 2 · GLUE IN · REFERENCE

Seafloor ages and color key

Distance from ridge (km)	Crust age (million years)
0	0
500	20
1,000	40
1,500	60
2,000	80
2,500	100
3,000	120

Age (million years)	Color
0	red
20	orange
40	yellow
60	green
80	blue
100	purple
120	dark purple

UNIT 3.1 · DAY 2 · GLUE IN · WRITE ON IT

Seafloor strip

Color each box using the key on D2-A. The ridge is done for you.

3,000 km	2,500	2,000	1,500	1,000	500	RIDGE	500	1,000	1,500	2,000	2,500	3,000 km
						red						

As you move away from the ridge, the crust gets _____ .

UNIT 3.1 · DAY 2 · GLUE IN · WRITE ON IT

Graph grid: age vs. distance

Mark a dot in the box where the distance column meets the age row. Connect the dots. The first dot (0 km, 0 years) is done for you.

Age (million years)	0 km	500	1,000	1,500	2,000	2,500	3,000 km
120							
100							
80							
60							
40							
20							
0	•						

Rate = distance ÷ time = _____ km per million years = _____ cm per year. Age at 3,500 km: _____ million years.

UNIT 3.1 · DAY 2 · GLUE IN · REFERENCE AND WRITE ON IT

North America crust ages and strip

Color the oldest region dark brown and the youngest light yellow. Choose three colors in between. The Canadian Shield is done for you.

Region of North America	Direction and distance from the central core (km)	Age of the crust
Pacific Coast Ranges	west, about 2,800	about 50 million years
Rocky Mountain region	west, about 1,800	about 150 million years
Canadian Shield (the central core)	0	about 2.7 billion years
Grenville Province (Great Lakes to Quebec)	east, about 1,200	about 1.1 billion years
Appalachian Mountains	east, about 2,000	about 400 million years

Pacific Coast Ranges (west)	Rocky Mountains	Canadian Shield (core)	Grenville Province	Appalachians (east)
		dark brown		

UNIT 3.1 · DAY 3 · GLUE IN · WRITE ON IT

Crater counts

Count the craters at each time step in the simulation. Row 1 is done for you. Then answer the questions.

Time step	Moon craters	Earth craters
1 billion years	6	5
2 billion years		
3 billion years		
4 billion years		

Which surface looks older? How can you tell?

Why does Earth show so few craters?

A surface has a lot of craters. Is it old or young? Why?

UNIT 3.1 · DAY 3 · GLUE IN · REFERENCE AND WRITE ON IT

Timeline table and old materials

Scale: 1 cm = 100 million years. Position = (4,500 − age in millions of years) ÷ 100. Two rows are done for you.

Event	Age (millions of years ago)	Position on the 45 cm strip (cm)
Earth forms (left end)	4,500	0
Oldest mineral on Earth (zircon)	4,400	1
Oldest known rock	4,000	
Oldest fossils	3,500	
Oldest ocean floor	200	
Today (right end)	0	45

Material	Where	Age
Meteorite fragments (Canyon Diablo)	Arizona, USA	about 4.56 billion years
Moon rocks from the lunar highlands	Brought back by Apollo	up to about 4.4 billion years
Oldest Earth mineral (zircon)	Jack Hills, Australia	about 4.4 billion years
Oldest known Earth rock (Acasta Gneiss)	Canada	about 4.0 billion years
Oldest widely accepted fossils (stromatolites)	Western Australia	about 3.5 billion years
Oldest ocean floor	Western Pacific and Atlantic margins	about 180–200 million years

UNIT 3.1 · DAY 4 · GLUE IN · WRITE ON IT

Guided notes

Fill in the blanks as your teacher goes. Blank 1 is done for you.

1. Old ocean crust is made at a **mid-ocean ridge** and moves away about 2.5 cm a year (25 km per million years).
2. A piece of crust travels 3,000 km. Time = 3,000 ÷ 25 = _____ million years. In 200 million years it travels 200 × 25 = _____ km, so it reaches a trench and sinks. No ocean floor is older than about _____ million years.
3. Ocean crust has a density of about **3.0 g/cm³**. Continental crust is about **2.7 g/cm³**. The _____ plate sinks (*subduction*).
4. Each continent has an ancient core. Volcanic arcs and pieces of crust are added to its edge over time, like rings on a tree. The edge is _____ than the core.

UNIT 3.1 · DAY 4 · GLUE IN · WRITE ON IT

Density table

Water has a density of 1.0 g/cm³. Density = mass ÷ volume.

Block	Mass (g)	Volume (cm³)	Density (g/cm³) = mass ÷ volume	Sinks or floats?
Wood block (stand-in for a continent)				
Clay block (stand-in for an old, cold ocean plate)				

Which block is denser? _____ What does the denser one do when they meet? _____

UNIT 3.1 · DAY 4 · GLUE IN · WRITE ON IT

SketchNotes checklist

Check each box when it is in your SketchNotes. Turn in a photo on Canvas: Lesson 4: SketchNotes.

	Half page of notes		Cross-section drawing
☐	Where ocean crust is made	☐	Ridge and plate moving away
☐	How it moves and sinks	☐	Trench and volcanic arc
☐	How continents grow from a core	☐	Continent with an ancient core and a younger edge
☐	Density numbers	☐	Density labels and a carbon arrow

UNIT 3.1 · DAY 5 · GLUE IN · REFERENCE

Magnetic stripe bands

Color both strips with this table (2 cm = 1 million years). Start at the end you will push through first.

Band (from the end you push first)	Color	Length	Age it represents
1	Reversed (blue)	3.4 cm	about 3.6–5.3 million years
2	Normal (red)	2.0 cm	about 2.6–3.6 million years
3	Reversed (blue)	3.6 cm	about 0.8–2.6 million years
4	Normal (red)	1.6 cm	0–0.8 million years (youngest)

UNIT 3.1 · DAY 5 · GLUE IN · REFERENCE AND WRITE ON IT

Hawaii and GPS

1 km per million years = 0.1 cm per year. GPS today: about 7 to 8 cm per year. The first rate is worked for you.

Volcano or island	Distance northwest of Kīlauea (km)	Age of its oldest rock (million years)
Kīlauea (Hawai'i Island, active now)	0	0
Kaua'i	about 500	about 5
Midway Atoll	about 2,500	about 28

Kīlauea to Kaua'i: 500 ÷ 5 = **100** km per million years = **10** cm per year (done for you).

Kīlauea to Midway: _____ ÷ _____ = _____ km per million years = _____ cm per year.

UNIT 3.1 · DAY 5 · GLUE IN · REFERENCE

Fossil table

Organism	What it was	Found in
Mesosaurus	freshwater reptile	South America and southern Africa
Glossopteris	seed fern (heavy seeds)	South America, Africa, India, Australia, Antarctica
Lystrosaurus	land animal	Africa, India, Antarctica

UNIT 3.1 · DAY 5 · GLUE IN · WRITE ON IT

Evaluate the evidence

Rate strength from 1 (weak) to 4 (very strong). Row 1 is started for you.

Line of evidence	Measured numbers?	Past or present motion?	Independent of the others?	Strength (1–4)
Rock ages across a ridge	Yes	Past		
Magnetic stripes				
GPS and Hawaiian island ages				
Matching fossils and rocks				

UNIT 3.1 · DAY 6 · GLUE IN · WRITE ON IT

Three buckets

Put each row of your Unit CER log in a bucket. One example is done for you.

The seafloor is young because…	The continents are old because…	The early record is missing because…
The oldest ocean floor is only about 200 million years old (done for you).		

UNIT 3.1 · DAY 6 · GLUE IN · REFERENCE

Part A sentence frames

1. **Claim:** Plate tectonics explains the ages of Earth's crustal rocks (very well / partly / poorly) because _____.
2. **Ocean evidence:** The seafloor is youngest at the _____ and gets older with _____; the oldest ocean floor is only about _____ million years old.
3. **Continent evidence:** Continental crust as old as _____ exists, and in North America the crust gets _____ toward the edges.
4. **Reasoning:** Because ocean plates are _____ than continental crust, they _____ at subduction zones, while continents _____.
5. **Limit or counterclaim:** One thing this evidence cannot show is _____. OR: Someone might say the ocean floor is young because _____. This does not fit the evidence because _____.

UNIT 3.1 · DAY 6 · GLUE IN · WRITE ON IT

Peer feedback

Trade papers. Give one star (something strong) and one step (one thing to improve).

Partner's name	One star	One step

UNIT 3.1 · DAY 6 · GLUE IN · WRITE ON IT

Evidence cards

Cut the cards apart. Put them in order from oldest to newest on a paper strip and glue them down. Card 1 is placed for you.

Evidence card	What it shows
Meteorites: about 4.56 billion years	The solar system formed about 4.56 billion years ago.
Moon highland rocks: up to about 4.4 billion years; heavily cratered	Solid crust existed early; the early solar system had many impacts.
Earth's oldest zircons: about 4.4 billion years	Solid crust and minerals had formed on Earth by then.
Oldest Earth rock (Acasta Gneiss): about 4.0 billion years	Rock has survived for 4 billion years in the continents.
Oldest fossils (stromatolites): about 3.5 billion years	Life existed by then.
Oldest ocean floor: about 200 million years	Ocean crust is recycled; the early record is not on the ocean floor.

UNIT 3.1 · DAY 6 · GLUE IN · WRITE ON IT

Part B story frames

Use your ordered evidence cards. Frame 1 is done for you.

1. About **4.56** billion years ago, **the solar system formed**. The evidence is **meteorites**.
2. About _____ billion years ago, _____ happened. The evidence is _____.
3. About _____ billion years ago, _____ happened. The evidence is _____.
4. Earth's earliest record is missing because _____.

UNIT 3.1 · DAY 7 · GLUE IN · WRITE ON IT

Self-rating chart

Rate each target from 1 (I need help) to 4 (I could teach it). Use your CFA scores to check.

Target	My rating (1–4)	My CFA score	What I will do today
ESS1-5.1: I can describe Earth's outer layer as moving plates, name three kinds of plate boundaries, and describe where ocean crust is made and where it sinks back down.			
ESS1-5.2: I can read a rock-age map and describe the pattern in the ages.			
ESS1-5.3: I can explain why ocean crust is young and why continents are oldest at their ancient cores.			
ESS1-5.4: I can judge how strong the evidence is that the crust has moved and is still moving.			
ESS1-5.5: I can judge how well plate tectonics explains the ages of rocks, using evidence.			